Mirna Ashri

mirnamernanni@gmail.com | m.ashri@campus.fct.unl.pt | mirna.ashri@etu.unistra.fr | LinkedIn | GitHub | +351 963328733

SUMMARY

Erasmus Mundus Joint Master's student in Chemoinformatics+ (NOVA University Lisbon & University of Strasbourg) with a BSc in Pharmaceutical Sciences (GPA 3.96/4.00, ranked 7th of 850). Background in quantum chemistry, DFT calculations and physical organic chemistry, combined with hands-on machine-learning experience: QSAR/QSPR modelling, representation comparison, validation and applicability-domain analysis, and Python workflows for atomistic and molecular data. Experimental training in synthesis and characterisation (NMR, ATR-FTIR, DSC) helps me connect computed quantities to measurable behaviour.

EDUCATION & FELLOWSHIPS

ChEmoInformatics+ Erasmus Mundus Joint Master - Lisbon-Strasbourg Track Sep 2025– Jun 2027
(NOVA University Lisbon & University of Strasbourg) Specialized in Organic Chemistry

- Coursework: Quantum Chemistry, Physical Organic Chemistry, Computational Chemistry, Molecular Modelling, Machine Learning, Organic Chemistry, Organic Synthesis, Medicinal Chemistry, Analytical Bio-organic Chemistry.
- Erasmus Mundus Joint Master Scholarship.

BSc in Pharmaceutical Sciences - Mansoura University, Egypt Sep 2018 – Aug 2023

- GPA: 3.96/4.00 | Ranked 7th among 850 students
- Full tuition waiver during the final three years.

RESEARCH EXPERIENCE

- **Research Intern – Computer-Aided Drug Design (CADD), LAQV-REQUIMTE, NOVA FCT (May 2026 – Present)**
  - Develop QSAR and machine-learning models to predict molecular activity within a computer-aided molecular design project.
  - Compare molecular descriptors, fingerprints and representations, and analyse chemical space to understand how representation shapes what a model can learn.
  - Assess model reliability through validation and applicability-domain analysis, i.e. identifying where predictions can and cannot be trusted.
  - Run virtual screening and molecular docking to study molecular interactions and prioritise candidate compounds.
  - Build reproducible Python/RDKit workflows for data preparation, modelling, validation and analysis.

- **Research Intern – Pharmaceutical Ionic Liquids, LAQV-REQUIMTE, NOVA FCT (June 2026 – July 2026)**
  - Synthesized and characterized pharmaceutical ionic liquids based on ammonium/morpholinium cations and carboxylate/amino-acid anions.
  - Applied NMR, ATR-FTIR, and DSC to characterize molecular structure and physicochemical properties.
  - Investigated molecular systems designed to modify the solubility and transport behaviour of poorly water-soluble compounds.
  - Connected experimental characterization with molecular structure and physicochemical behaviour.
- **Planned Master's Thesis - Computational Fragment Screening, University of Münster, Germany - PLANNED (Jan 2027 – Jun 2027)**
  - Computational and analytical investigation of a fragment library developed for affinity-mass-spectrometry applications.
  - Planned work includes screening molecular fragments and analysing their interactions with target systems.
  - Research direction combines analytical chemistry, molecular structure analysis, computational screening, and mechanistic interpretation.

PROJECTS

- **Aqueous Solubility Prediction Using Hybrid Molecular Representations | *Python, RDKit, scikit-learn, XGBoost, Random Forest, Streamlit***
  - Developed machine-learning models for aqueous solubility prediction using approximately 9,980 compounds.
  - Combined Morgan molecular fingerprints with physicochemical descriptors to investigate the effect of molecular representation on prediction.
  - Trained and evaluated Random Forest and XGBoost models and analysed structure–property relationships.
  - Developed a Streamlit application for molecular property prediction from SMILES input.
  - Presented the project as a poster at the 2026 Strasbourg Summer School on Chemoinformatics.

- **Molecular Data Machine Learning, Kaggle competition (MMLI / UIUC) | Python, RDKit, scikit-learn, QSAR, Feature Selection, Symbolic Regression**
  - Predicted experimental photostability lifetimes (T80) from ~150 molecular features using Ridge regression and Random Forest.
  - Assessed generalisation with repeated K-fold cross-validation and MSLE; compared full and reduced feature sets (SelectKBest).

- Investigated feature relevance with correlation analysis, permutation importance, Lasso and symbolic regression to obtain interpretable structure–property relationships.

- **Enveda CASMI 2026: Molecule ID From Mass Spectra (Kaggle) | Python, RDKit, LC-MS/MS, Machine Learning, Molecular Structure Prediction (In Progress)**
- Machine-learning models for predicting molecular structures from tandem mass spectra.

PROFESSIONAL EXPERIENCE

Faculty of Pharmaceutical Sciences, Mansoura University, Mansoura, Egypt
Teaching and Research Assistant of Analytical Chemistry January 2024 – Present

- Contributed to research in pharmaceutical and analytical chemistry, supporting experimental design, data analysis, and interpretation of research findings.
- Applied analytical methods for the characterization of pharmaceutical and biologically relevant compounds.
- Supported undergraduate teaching and mentoring in analytical chemistry, fostering interdisciplinary scientific thinking and problem-solving.

Bringing Europeans Together Association - BETA Europe Remote, Belgium
Vice-President Aug 2024 – June 2026
• Led an international team of volunteers in a European association, working across countries and time zones.
Model European Union Strasbourg (MEUS), BETA France Remote, France
Diversity Coordinator, Ex-Secretary-General, and Ex-Partnerships Manager Feb 2022 – Nov 2024
• Held successive responsibilities coordinating an international event team, managing external partnerships and overseeing the organisation's operations.
Oath for Sustainability Network - Climate and Health Research Lab Remote, Africa
Research Fellow Feb – June 2024

LANGUAGES

- **Arabic – Native**
- **English – Advanced C1, French – Advanced C1, German – Advanced C1**
- **Spanish – Intermediate B1, Italian – Intermediate B1**
- **Turkish – Basic A2, Portuguese - Basic A2**
- **Chinese (Mandarin) – Basic HSK1**

SOFTWARE PROFICIENCY

- **Computational Chemistry Softwares**
  DFT (ORCA, MOPAC), molecular modelling, docking (AutoDock Vina),
  molecular dynamics (AMBER), QSAR/QSPR, virtual screening,
  chemical-space analysis; GaussView, PyMOL, Open Babel, DataWarrior, YASARA.
- **Computational Chemistry**
  Molecular modelling, Molecular docking, Molecular Dynamics,
  Molecular descriptors and fingerprints, Molecular similarity, Chemical-space analysis,
  DFT Calculations,
  Applicability-domain analysis, Virtual screening, Structure-property relationship analysis
- **Programming & Data Science**
  Python, RDKit, NumPy, Pandas, SciPy, scikit-learn
  XGBoost, Matplotlib, Jupyter, SQL, Bash
- **Experimental Chemistry**
  organic synthesis and purification of ionic liquids; NMR, ATR-FTIR, DSC,
  Spectrofluorimetry, pharmaceutical analytical chemistry.
  Coursework knowledge of HRMS, HPLC and X-ray crystallography.
- **Tools & Development**
  GitHub, Linux, Conda, Streamlit

AWARDS & RECOGNITIONS

- **Ideal Student Competition - All Egyptian Universities - 2022 (Third Place)**
- **Al-Abakera Families Program Show - 2020 - Season 2 (First Place)**
- **Poster presenter, 2026 Strasbourg Summer School on Chemoinformatics.**

PODCASTS

MEUStream Podcast Series – (Host) Podcasting with Members of the European Parliament
Coucou De France: (Guest) My Journey with the French Language – Episode II (19 September 2024)
Radio La Voce Di Arda: (Host) Tolkien Course – Episode XIX (29 July 2022) - Podcast Feature on Apple Podcasts.